

Knowledge and Reasoning

A good chess program can defeat human masters. However it does not know that a chess board can be used for playing checkers as well. It cannot decide when to play a chess game.

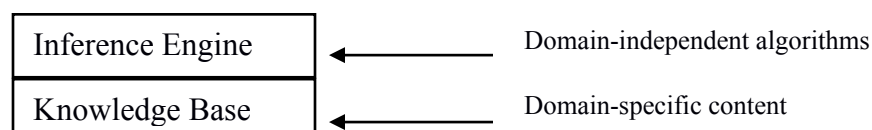
Reasoning is defined as the use of reason, especially in drawing of inferences or conclusions (Merriam-Webster Dictionary). Reason is the power of comprehending, inferring or thinking especially in orderly rational ways (Merriam-Webster Dictionary).

Humans have knowledge about things and can do reasoning. It is also important for an artificial being, which we are calling an **agent**, to achieve “high level” intelligence.

Knowledge and Reasoning

- enable agents to cope with complex environments.
 - *[E.g., how does one schedule an around-the-world trip? One needs the knowledge about “the world”.]*
- play a crucial role in dealing with partially observable environments.
 - *[Part of states could be hidden. E.g., how does one avoid collisions in a busy shopping mall?]*
- enable agents to handle complicated tasks, e.g., understanding natural language.
 - *[E.g., "Mike opened the door, found a chair and sat on it". What does ‘It’ refer to? Door or chair?]*
- provide more flexibility in problem solving.
 - *[Decisions are based on updated knowledge. E.g., how does one choose a good text book for a course?]*

To be intelligent, an agent needs a **knowledge base** to store knowledge and an **inference engine** to do the reasoning.



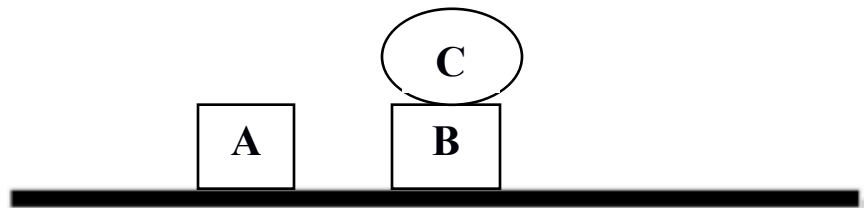
Knowledge Representation

An agent must be able to

- represent knowledge of the world, including states, actions etc.
- incorporate new percepts.
- update internal representations of the world.
- deduce hidden properties of the world
- deduce appropriate actions.

It is desirable to have a Knowledge Representation (KR) scheme with the following properties:

- **Expressive:** It must be able to represent as much as possible about the world.
- **Precise:** It must be clear and unambiguous.
 - An example of ambiguity: *The man saw the boy in the park with the telescope.*
 - Whose telescope? The man's or the boy's?
 - Could it be that there was a park with a telescope as against other parks without telescopes?
 - Did the man use the telescope to see the boy in the park?
- **Regular:** There must be a clear mapping between the knowledge and its representation.
- **Adaptable:** One must be able to add new information or delete invalid information.
- **Suitable for reasoning:** New knowledge can be inferred
- **Computationally attractive:** The scheme should be able to be implemented.
- **Meta level Reasoning:** It can represent knowledge about knowledge
- **Qualitative:** it can represent qualitative knowledge. For example



How does one represent “you can only put block A on block B if B is clear?”

Physical Symbol Hypothesis

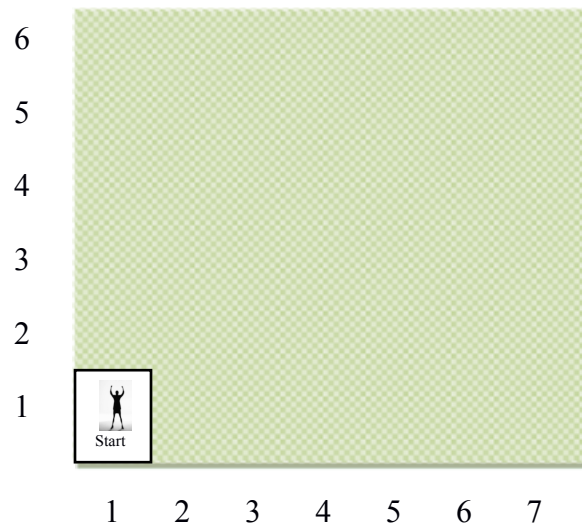
The following are **necessary** and **sufficient** conditions for intelligent behaviour:

1. There should be symbolic patterns for the representation of significant aspects of a problem domain.
2. There should be operations or processes on these patterns which are used to generate new patterns which are potential solutions to problems.
3. A search should be carried out to select a solution from among these possibilities.

The physical symbol hypothesis is the foundation of all work in (Symbolic) Artificial Intelligence.

Example: The Wumpus World

Goal: The hunter must find the gold without being eaten by a wumpus or falling in a pit.



In this world

- Squares adjacent to the wumpus are smelly
- Squares adjacent to the pit are breezy
- There is a glitter if gold is in the same square.
- The hunter can only perceive one square.

To solve the wumpus world problem, an “intelligent” hunter should:

- Use symbolic patterns to represent the information about wumpus world, such as “squares adjacent to wumpus are smelly”, “I am in a breezy square”, etc.
- Define operations which can be used to search a solution. Such as “move left”, “move up”, etc.
- Perform a search to find the solution, in this case, reaching gold.

The Wumpus world is partially observable (i.e., only local perception).

Some information is **static** (i.e. holds true forever), such as

- *If there is a breeze in square (X , Y), then there is a pit in one of the adjacent squares.*

Other information is **dynamic**, (i.e. depends on the current state of the world)

- *There is a glitter at square (2 , 3).*
- *The wumpus is alive.*

If we tell the hunter the static information at the start of its life, and then at some later point tell it that there is a breeze at (3 , 2), the agent should be able to **automatically infer** that there is a pit in an adjacent square.

How? By using a powerful gadget from philosophers’ toolbox – **LOGIC**

Logic

Logic is the science of reasoning, proof, thinking and inference - *Concise Oxford English Dictionary*. It is an important formalism for knowledge representation. It is a powerful inference method.

Logic is a set of formal languages for representing information such that conclusions can be drawn. For example,

“ α is true so β is true” or “ β is true where α is true” is represented as

$$\alpha \models \beta$$

This expression is called entailment; one thing follows logically from another. α entails β . If α is true then β must also be true.

Symbol α could mean “ $X > Y$ ”, β could mean “ $Y < X$ ”. Then the above expression means “if $X > Y$ then $Y < X$ ”.

Q: Does that sound logical??

Components of a Logic System

There are three main components of a logic system:

- **Syntax:** Syntax defines legal expressions and sentences in the language.

In using the language of arithmetic as an analogy:

$x + 2 > 3$ is a legal sentence, but $x + 5 - <$ is not.

A grammatically correct sentence in **logic** is called **Well Formed Formula (WFF)**.

- **Semantics:** This defines the “meaning” of sentences – whether they hold the truth.

In using the language of arithmetic as an analogy:

$x + 2 > x$ is always true.

$3 - 1 = 0$ is always false.

- **Inference mechanism:** It is used for deciding what follows validly from a set of logical sentences.

Propositional Logic

Syntax

Propositional Logic (PL) is the simplest logical system. The basic symbols of PL are:

- **logical constants** (T or *True*, F or *False*)
- **Proposition symbols** (e.g., P , Q , R ,)
- **Connectives** ($\wedge$, $\vee$, $\neg$, $\rightarrow$, $\leftrightarrow$)

Connectives are used to build more complex sentences from simpler ones.

For example, if $P_{3,1}$ stands for “*There is a pit in square 3,1*” and $P_{3,2}$ stands for “*There is a pit in square 3,2*”, then

$P_{3,1} \vee P_{3,2}$ stands for “*There is a pit in square 3,1 or square 3,2*”

$P_{3,1} \wedge P_{3,2}$ stands for “*There is a pit in square 3,1 and square 3,2*”

Sentences in Propositional Logic

The logical constants ***True*** and ***false*** are sentences.

If P is a sentence, then (P) is a sentence.

If P and Q are sentences, then $P \wedge Q$ is a sentence (**Conjunction**).

If P and Q are sentences, then $P \vee Q$ is a sentence (**Disjunction**)

If P and Q are sentences, then $\neg P$ is a sentence (**Negation**).

If P and Q are sentences, then $P \rightarrow Q$ is a sentence (**Implication**)

If P and Q are sentences, then $P \leftrightarrow Q$ is a sentence (**bi-condition**).

A **literal** is an atomic sentence, either Positive P or negative $\neg P$

Examples of PL sentences

- P
- $P \wedge Q$
- $P \vee Q$
- $(P \vee Q) \rightarrow R$
- $((P \wedge Q) \rightarrow (\neg R \vee (S \wedge P)))$

Note: Parentheses must be used to avoid ambiguity.

The following sentences are **not** WWFs:

- $P\neg$
- $P \wedge \neg$
- $\wedge P Q$
- $\rightarrow P$

Semantics of Propositional Logic

Semantics define the “meaning” of sentences. The meaning of a logical sentence is based on whether it is true or false. Truth values of **proposition symbols** are assigned by **interpretations**, which correspond to the way the world could be, such as symbols.

$P_{3,1}$ refers to *There is a pit in square 3,1* (Wumpus world)

Q refers to $X > 3$

Depending on the situation, the above symbols could be true or false. Truth values of **WWFs** are worked out from truth values of components.

Whether a propositional symbol is true or not depends on its interpretation. Whether or not **P** is true depends on what **P** represents.

Some propositions are always true and some are always false. Some could be true or false, depending on the current state. Also, propositions are always true within a certain scope or world.

“*Squares adjacent to wumpus are smelly*” might not be true outside the wumpus world.

“*Elephants can fly*” might be true in a fantasy world.

Truth Value of Sentences

The truth value of a sentence is based on the truth values of its propositional symbols.

- $P \wedge Q$ is true if and only if (iff) P is true and Q is true
- $P \vee Q$ is true iff P is true or Q is true
- $\neg P$ is true, iff P is false.
- $P \rightarrow Q$ is true, iff P is false and Q is true, or P and Q are both true or false
- $P \leftrightarrow Q$ is true, iff P and Q are both true, or P and Q are both false

A truth table can be used to summarize the truth value of these simple sentences:

P	Q	$\neg P$	$P \wedge Q$	$P \vee Q$	$P \rightarrow Q$	$P \leftrightarrow Q$
false	false	true	false	false	true	true
false	true	true	false	true	true	false
true	false	false	false	true	false	false
true	true	false	true	true	true	true

Implication

The truth values of implication, $\rightarrow$, are somewhat counter-intuitive. Let P refer to “*He studied Hard*” and Q refer to “*He passed the exam*”.

So $P \rightarrow Q$ will read, “*He studied hard implies he passed the exam*”.

Therefore let’s check – are these statements true?

- True when P is true and Q is true – He studied hard so he passed the exam
- True when P is false and Q is true – He didn’t study hard so he didn’t pass the exam
- False when P is true and Q is false – he studied hard but he didn’t pass the exam
- True when P is false and Q is false – he didn’t study hard but he did pass the exam

The last one seems bizarre. It actually says if he passed the exam then the implication is true even if the premise/antecedent is false

“*Dogs love cats implies cats hate dogs*”. This statement is odd but logically true (if cats hate dogs is always true).

Truth Tables

For complex sentences, truth tables can be used to determine the truth values by determining the truth values of its components.

Example:

P	Q	R	$P \wedge Q$	$R \vee Q$	$(P \wedge Q) \rightarrow (R \vee Q)$
T	T	T	T	T	T
T	T	F	T	T	T
T	F	T	F	T	T
T	F	F	F	F	T
F	T	T	F	T	T
F	T	F	F	T	T
F	F	T	F	T	T
F	F	F	F	F	T

Wumpus World Sentences

Now let us go back to the wumpus world and represent the information of the wumpus world in PL.

Let $P_{i,j}$ be true if there is a pit in square $[i, j]$.

Let $B_{i,j}$ be true if there is a breeze in square $[i, j]$

$\neg P_{1,1}$ There is no pit in square $[1, 1]$.

$\neg B_{1,1}$ There is no breeze in square $[1, 1]$.

$B_{2,1}$ There is a breeze in square $[2, 1]$.

$B_{2,1} \leftrightarrow (P_{1,2} \vee P_{2,1})$ Pits cause breezes in adjacent squares.

$B_{2,1} \leftrightarrow (P_{1,1} \vee P_{2,2} \vee P_{3,1})$ Pits cause breezes in adjacent squares.

Based on these given information, we can infer that there is not pit in square $[1, 2]$. One may ask how this is possible. We will study some key concepts of logic before we go through inferencing processes.